

MATH 1342 – Elementary Statistics – Week 4: Percentiles, Boxplots and Two-Variable Relationships

Sep 15 – 17 | 2.3 One Quantitative Variable: Percentiles • 2.4 Two Variable Relationships | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 2. Calculate and interpret measures of central tendency and dispersion, including mean, median, standard deviation, and quartiles. 4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 5. Analyze relationships between two quantitative variables using correlation and linear regression.	1. Compute the sample standard deviation of a small dataset by hand. (CO2) 2. Compute and interpret z-scores to compare values measured on different scales or in different groups. (CO2, CO4) 3. Find the five-number summary, range, and IQR; apply the 1.5×IQR rule to identify outliers; and construct or interpret a boxplot. (CO1, CO2) 4. Compare a quantitative variable across groups with side-by-side boxplots, and compute and interpret a difference in means. (CO1, CO2) 5. Describe the direction, form, and strength of the relationship in a scatterplot and estimate the correlation r. (CO5) 6. Use technology to compute r and explain how outliers, nonlinear patterns, and the absence of causation limit its interpretation. (CO5)	e-Text: 2.3 One Quantitative Variable: Using Percentiles	MO1, MO2, MO3	Participation: One Quantitative Variable (Kahoot!) – Tue: review of 2.2	MO1	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles	MO1, MO2, MO3	Participation: Using Percentiles – Your Turn: comparing heights with z-scores	MO2, MO3		
		Slides: 2.3 One Quantitative Variable: Using Percentiles	MO1, MO2, MO3	Participation: Box Plots and Outliers – five-number summary, 1.5×IQR rule	MO3	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 20, 11:59 PM)	MO4, MO5, MO6
		e-Text: 2.4 Two Variable Relationships	MO4, MO5, MO6	Quick Self-Quizzes: tea vs. coffee boxplots and difference in means; NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier	MO4, MO5, MO6		
		Slides: 2.4 Two Variable Relationships	MO4, MO5, MO6			Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.4	All